kernel method Numerical

Sample $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}_{3\times 1}$

feature

$\Theta(x) = \begin{bmatrix} x_1, x_1 \\ x_1, x_2 \\ x_1, x_3 \\ x_2 \; x_1 \\ x_2 \; x_2 \\ x_2 \; x_3 \\ x_3 \; x_1 \\ x_3 \; x_2 \\ x_3 \; x_3 \end{bmatrix}_{9\times 1}$

$z = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}_{3\times 1}$

(feature map)

$\Theta(z) = \begin{bmatrix} z_1 \; z_1 \\ z_1 \; z_2 \\ z_1 \; z_3 \\ z_2 \; z_1 \\ z_2 \; z_2 \\ z_2 \; z_3 \\ z_3 \; z_1 \\ z_3 \; z_2 \\ z_3 \; z_3 \end{bmatrix}_{9\times 1}$

$\Theta(x) = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 2 \\ 4 \\ 6 \\ 3 \\ 6 \\ 9 \end{bmatrix}$ $\times$ $\Theta(z) = \begin{bmatrix} 16 \\ 20 \\ 24 \\ 20 \\ 25 \\ 30 \\ 24 \\ 30 \\ 36 \end{bmatrix} = \begin{bmatrix} \\ \\ \\ \\ \\ \\ \\ \\ \\ \end{bmatrix}$

$\Theta(x) \times \Theta(z) = 16 + 40 + 72 + 40 + 100 + 180 + 72 + 180$

$= 324 = \cancel{1030} = \boxed{1024}$

↳ this product is done in Higher dimension space

$\phi(x) \quad \phi(z)$

kernal trick

$\boxed{k(x,z) = (\langle x, z \rangle)^2}$ $\quad x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad z = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$

$k(x,z) = (x_1 z_1 + x_2 z_2 + x_3 z_3)^2$

$= (4 + 10 + 18)^2$

$= (32)^2 = \boxed{1024}$

Polynomial kernel of degree 2

kernel function :- [By doing Calculation on original Dimension Space] we get same value as higher dimension function

Calculation on original dimension → But Calculate for Higher dimension

	x_1	x_2	y
$x^{(1)}$	1	1	2
$x^{(2)}$	3	1	4

$x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ New test Sample

Hypothesis fun : $\boxed{h_\theta(x) = \theta^T \phi(x)}$

$\theta \in \mathbb{R}^p$ $\quad \phi(x) \in \mathbb{R}^p$

$$h_\theta(x) = \sum_{i=1}^{n} \beta_i \, k(x^{(i)}, x)$$

* assuming β_i learned from training procedure

we need all β's which are 2 in this dataset as 2 sample

let us assumes $\beta_1 = 1$ $\quad \beta_2 = 1$

$e^{-\|x-z\|^2}$

kernel = Radial Basics function (RBF)

$$k(x, z) = \exp(-\|x - z\|^2$$

where

$$\|x - z\|^2 = (x_1 - z_1)^2 + (x_2 - z_2)^2$$

Solving

$$= \sum_{i=1}^{n} \beta_i \, k(x^{(i)}, x)$$

$$= \beta_1 k(x^{(1)}, x) + \beta_2 k(x^{(2)}, x)$$

$$= \beta_1 \exp(-\|x^{(1)} - x\|^2) + \beta_2 \exp(-\|x^{(2)} - x\|^2)$$

Compute;

(i) $\|x^{(1)} - x\|^2$

$$= (x_1^{(1)} - x_1)^2 + (x_2^{(1)} - x_2)^2$$

$$= (1-2)^2 + (1-1)^2$$

$$= (-1)^2 + 0$$

$$= 1$$

ii) $\|x^{(2)} - x\|^2$

$$= (x_1^{(2)} - x_2)^2 + (x_2^{(2)} - x_2)^2$$

$$= (3-2)^2 + (1-1)^2$$

$$(1)^2 \qquad 0$$

$$= 1$$

Substituting β values

$$k(x^{(1)}, x) = \exp(-\|x^{(1)} - x\|^2)$$
$$= \exp(-1)$$
$$\boxed{= 0.367}$$

$$k(x^{(2)}, x) = \exp(-\|x^{(2)} - x\|^2)$$
$$= \exp(-1)$$
$$\boxed{= 0.367}$$

$$h_\theta(x) = \beta_1 k(x^{(1)}, x) + \beta_2 k(x^{(2)}, x)$$
$$= 1 \times 0.367 + 1 \times 0.367$$
$$h_\theta(x) = \boxed{0.734}$$ this is the final prediction we get